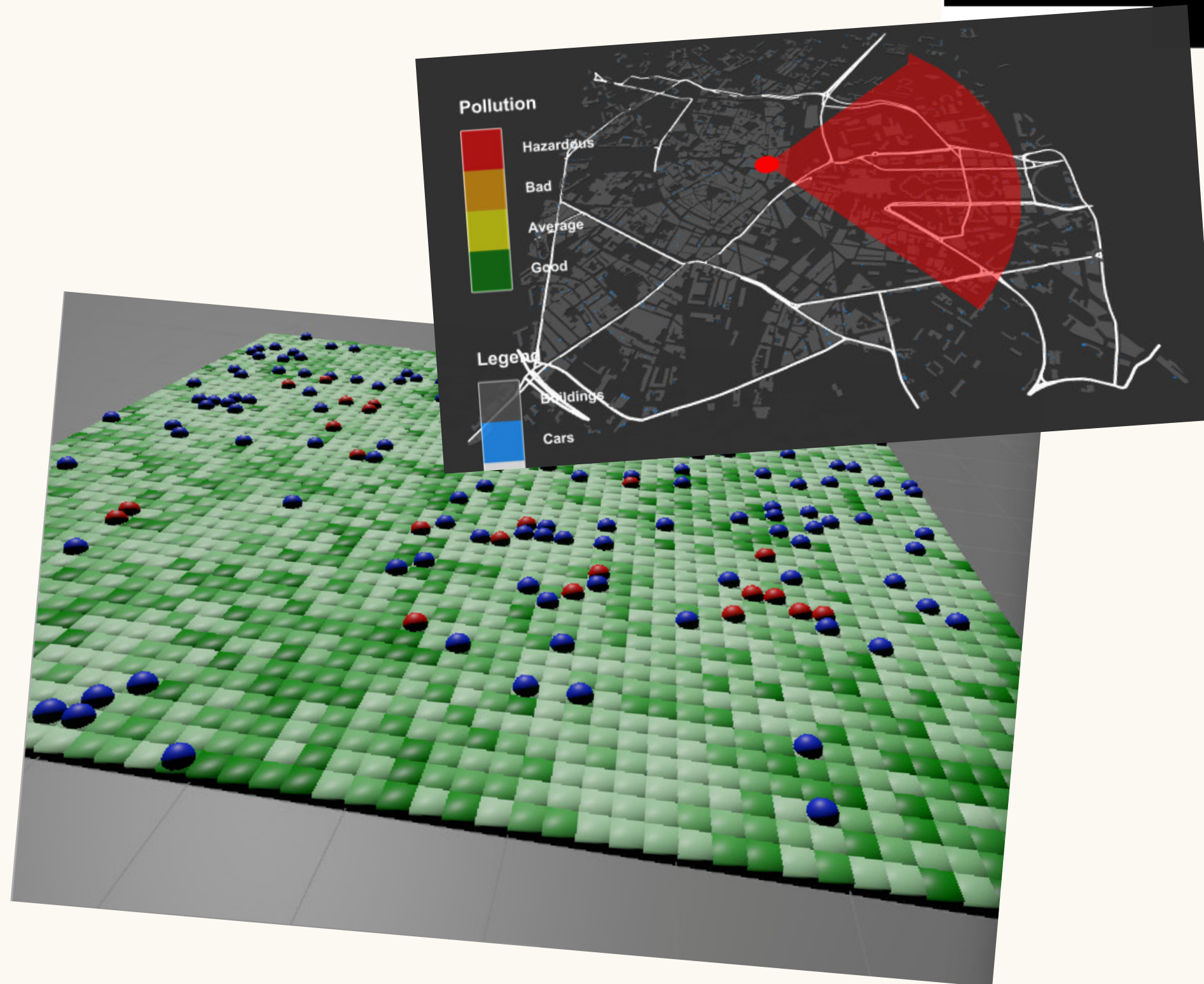
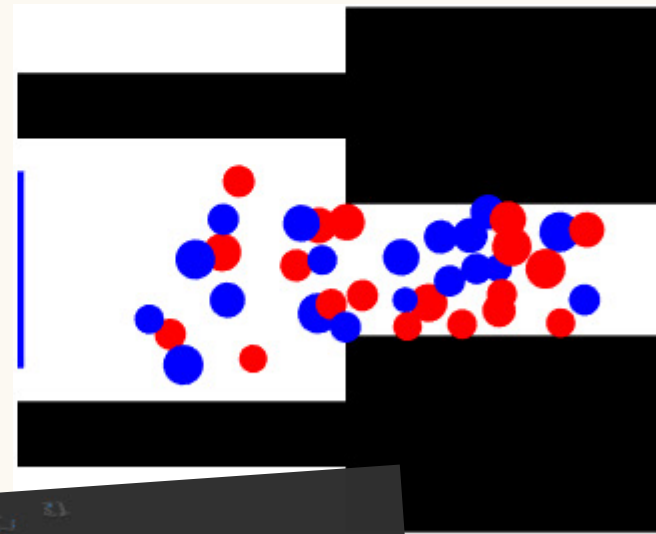


Day 2 GAMA-Unity Workflow

Understanding and mastering the SIMPLE workflow



Outline



1. Overview of the Workflow
2. Setup Requirements and Preparation
3. Unity Package and Scene Setup
4. Creating and Exporting the Experiment
5. First Live Run and Verification
6. Visual Personalization
7. Static Preview Usage
8. Dynamic Color Integration
9. Afternoon Mission and Guided Lab

Overview of the Workflow

THE APPLICATIONS

Three tools, three roles



Unity

builds the 3D experience users can see and interact with.



SIMPLE.Middleware

opens a websocket connection between GAMA and Unity.



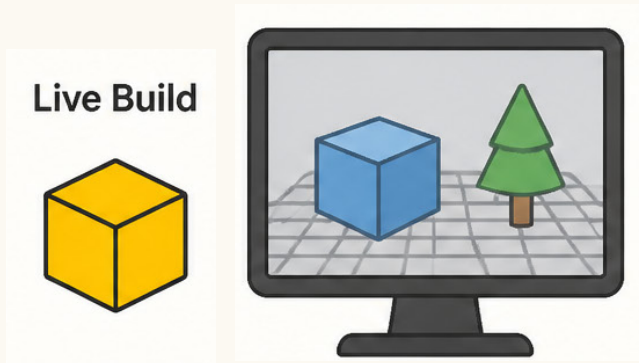
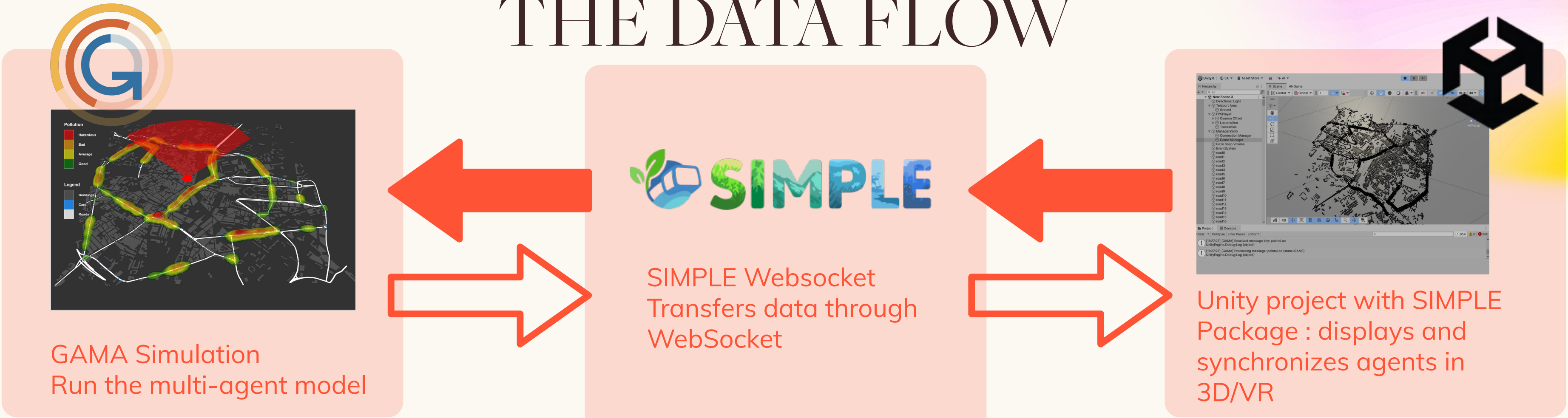
GAMA

runs the simulation and updates the agents' behavior.

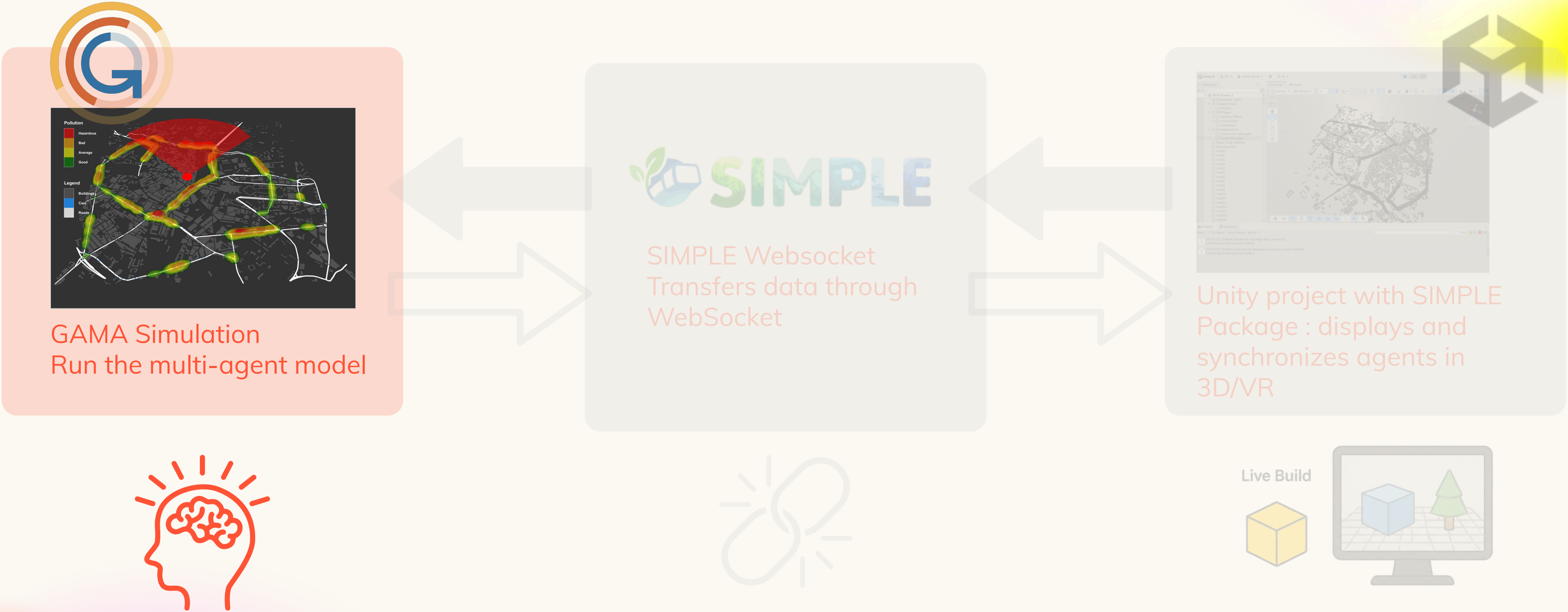
Why we connect them ?

The SIMPLE workflow links GAMA decisions to Unity visuals :
we link the **simulation** possibilities of **GAMA** with the
rendering possibilities of **Unity**.

THE DATA FLOW



GAMA Setup

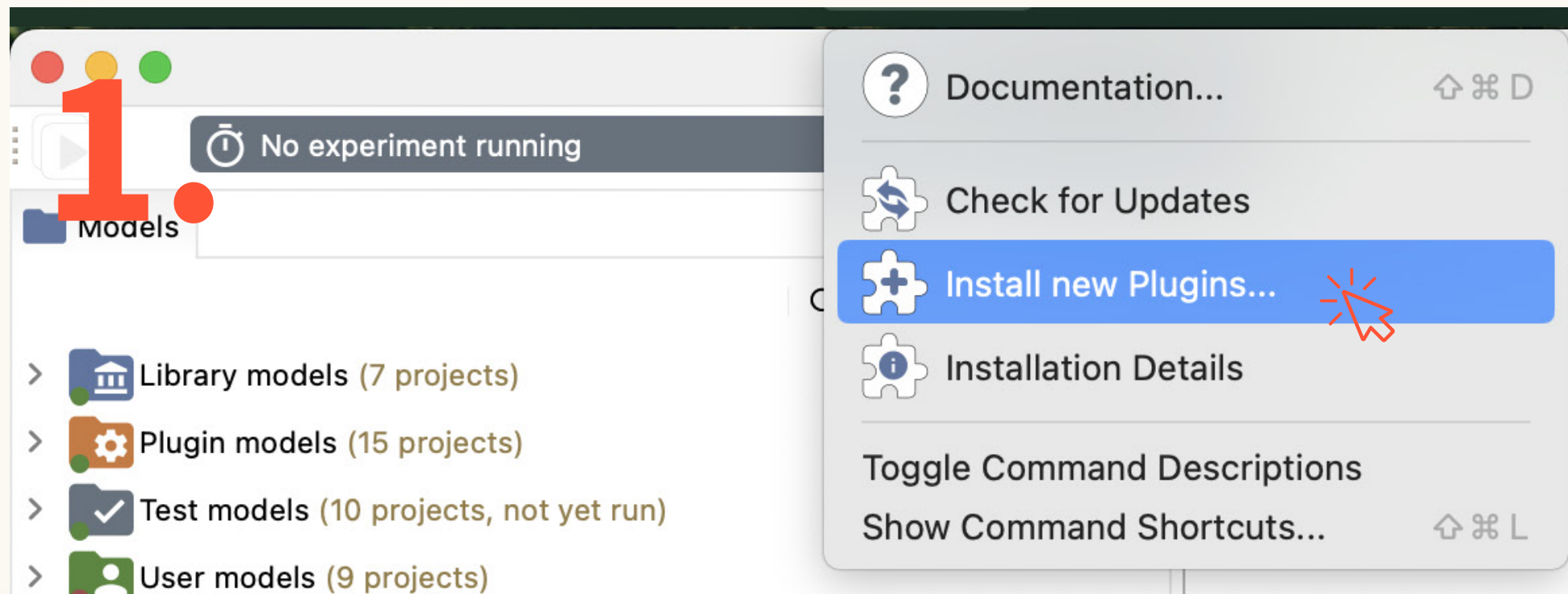


GAMA Plugin



The **SIMPLE Unity plugin** converts a standard GAMA experiment into a **Unity-compatible "vr_xp"** experiment, configuring the middleware messages required to run and synchronize the simulation with Unity.

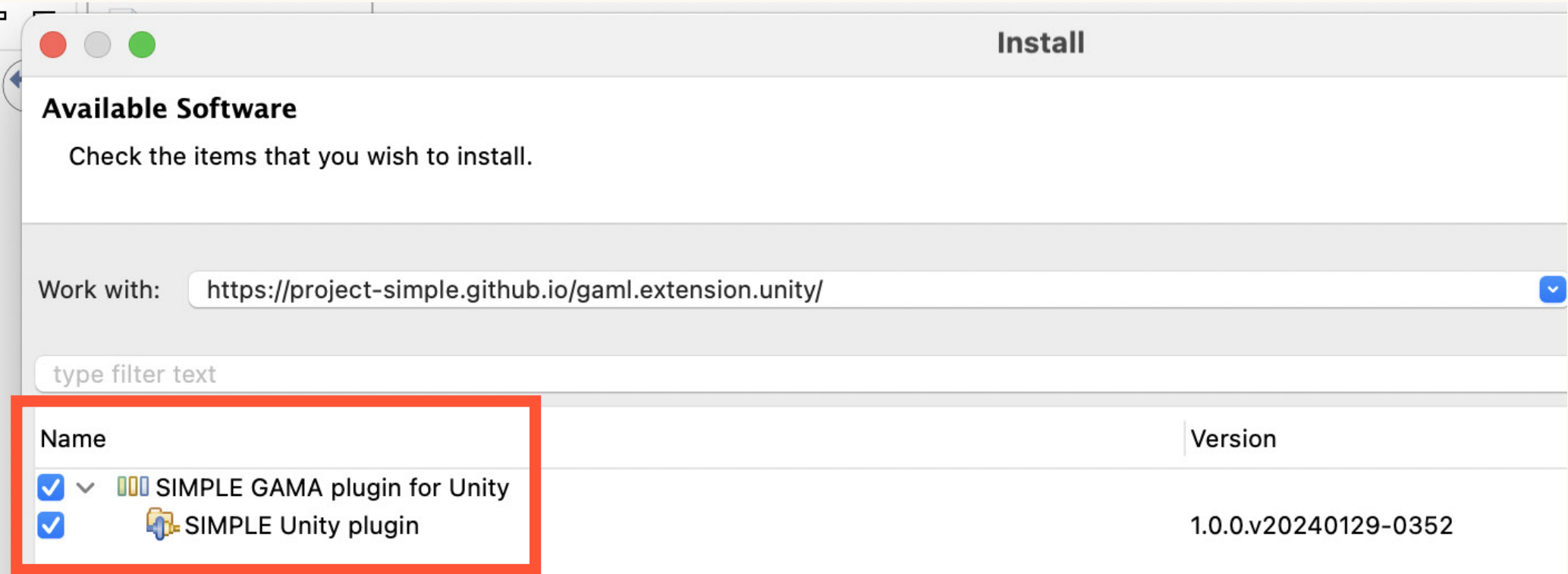
—————> The SIMPLE Unity Plugin installation : <https://doc.project-simple.eu/gama/installation>



2. In the dialog, click Work **with** and enter this update site URL:

<https://project-simple.github.io/simple.toolchain/>

3. Press Enter. The plugin list loads.



4. Select "SIMPLE Unity plugin", click "Next" and finish installation

Creating and Exporting the Experiment

The GAMA Experiment

It is time to run the first GAMA experiment in Unity with this package. For this tutorial, use the **6th prey Predator** model located in the following hierarchy from GAMA :

Library models > Tutorials > Predator Prey > models > Model 06.gaml

⚠ Warning

The original GAMA experiment cannot be used directly in Unity. It must first be converted into a `vr_xp` experiment with the SIMPLE Unity plugin.

Open the "**Prey Predator**" model in the editor, then, select in the "**Unity VR**" menu "**Model to VR**". A Wizard should open. The "**Unity VR**" menu only appears if you have the **mouse cursor in the text editor** of the model.



2.1 Convert the GAMA Experiment to `vr_xp`

Before running the model in Unity, convert the GAMA experiment with the SIMPLE Unity plugin. During this conversion, each species that should appear in Unity must be explicitly exported.

On the first **Definition of the VR experiment** screen, keep the default values and click **Next**.

Definition of the VR experiment

Enter information to build the VR experiment

Minimum duration of a cycle in seconds (minimum_cycle_duration):

General parameters

Perception radius of players

Min distance between agents to send to be considered

Displays

Main display:

Displays to hide:

☒ main_display ☒ info_display

☒ Number of preys ☒ Number of predators



VR Experiment generation

Export species

Specify the species to export to Unity, and their aspect in Unity

Species

☒ prey (exported, dynamic)

☐ predator (not exported)

☐ vegetation_cell (not exported)

☐ generic_species (not exported)

☒ Update every

1

simulation cycles

Aspect in Unity

default

Uses ☐ A prefab defined in Unity ☒ The geometry of the agent in GAMA

Color:

Material:

Height/depth:

1.0

Buffer:

0.0

☐ has a collider

☐ is interactable

☐ synchronize with GAMA

☐ is grabable

Tag:

+

-

?

< Back

Next >

Finish

Cancel

On the **Export species** screen, do not immediately click **Next**. Select a species on the left, then click the **+** button under **Aspect in Unity**.

VR Experiment generation

Export species

Specify the species to export to Unity, and their aspect in Unity

Species

- ☒ prey (exported, dynamic)
- ☐ predator (not exported)
- ☐ vegetation_cell (not exported)
- ☐ generic_species (not exported)

☒ Update every simulation cycles

Name of the Unity properties

Enter a name for the set of properties used in Unity

Keep the **default** property name and click **OK**.

OK Cancel

+ - Tag:

< Back Next > Finish Cancel

Same for each species :
prey
predator
vegetation_cell

VR Experiment generation

Export species

Specify the species to export to Unity, and their aspect in Unity

Species

☒ prey (exported, dynamic)

☒ predator (exported, dynamic)

☒ vegetation_cell (exported, dynamic)

☐ generic_species (not exported)

Update every

1

simulation cycles

Aspect in Unity

default

prey

predator

vegetation_cell

+

-

Uses ☐ A prefab defined in Unity ☒ The geometry of the agent in GAMA

Color:

Material:

Height/depth:

1.0

Buffer:

0.0

☐ has a collider

☐ is interactable

☐ synchronize with GAMA

☐ is grabable

Tag:

?

< Back

Next >

Finish

Cancel

At the end, these species should be marked as exported.

Click Next

The c

What you should see now :

The image shows a software interface with two main panels. The left panel is a file explorer titled 'Models' with a search bar 'Find model...'. It displays a folder structure: 'Lunera flu (7 models)', 'Predator Prey (14 models)', and 'models (14 models)'. Under 'models', there are files 'Model 01.gaml' through 'Model 06.gaml' and 'Model 06-VR.gaml'. 'Model 06-VR.gaml' is highlighted with a red box. A red arrow points to it with the text '1. A brand new VR model'. The right panel is a code editor showing a GAML script for a prey-predator model. At the top, there are tabs for 'Traffic and Pollut...' and 'Traffic and...'. Below the tabs, there are buttons: a green '+' button, a green 'prey_predator' button, and a green 'vr_xp' button. These three buttons are enclosed in a red box. A red arrow points to the '+' button with the text '2. vr-xp running option'. The code editor shows the following GAML script:

```
1 model prey_predator_m
2
3 import "Model 06.gaml
4
5 species unity_linker
6   This definition
7   string player_spec
8   This definition
9   int max_num_player
10  unity_property up
11  unity_property up
12  unity_property up
13  unity_property up
14  This definition
15  list<point> init_
```

- After this, click **Next**, keep the default values, set the number of players between **0** and **1**, then click **Finish**.
- For more details, this step is explained [in this tutorial](#)

 Important

If the species are not added on the **Export species** screen, the experiment may still start in GAMA and the Unity player may be created, but no simulation agents will appear in Unity.

- The **generic_species** entry is abstract and is not required for the visual result. Exporting it is harmless, but it can also be ignored.

Our experiment is ready to connect to the Middleware !



CREATE THE `vr_xp` EXPERIMENT

Generate the VR experiment

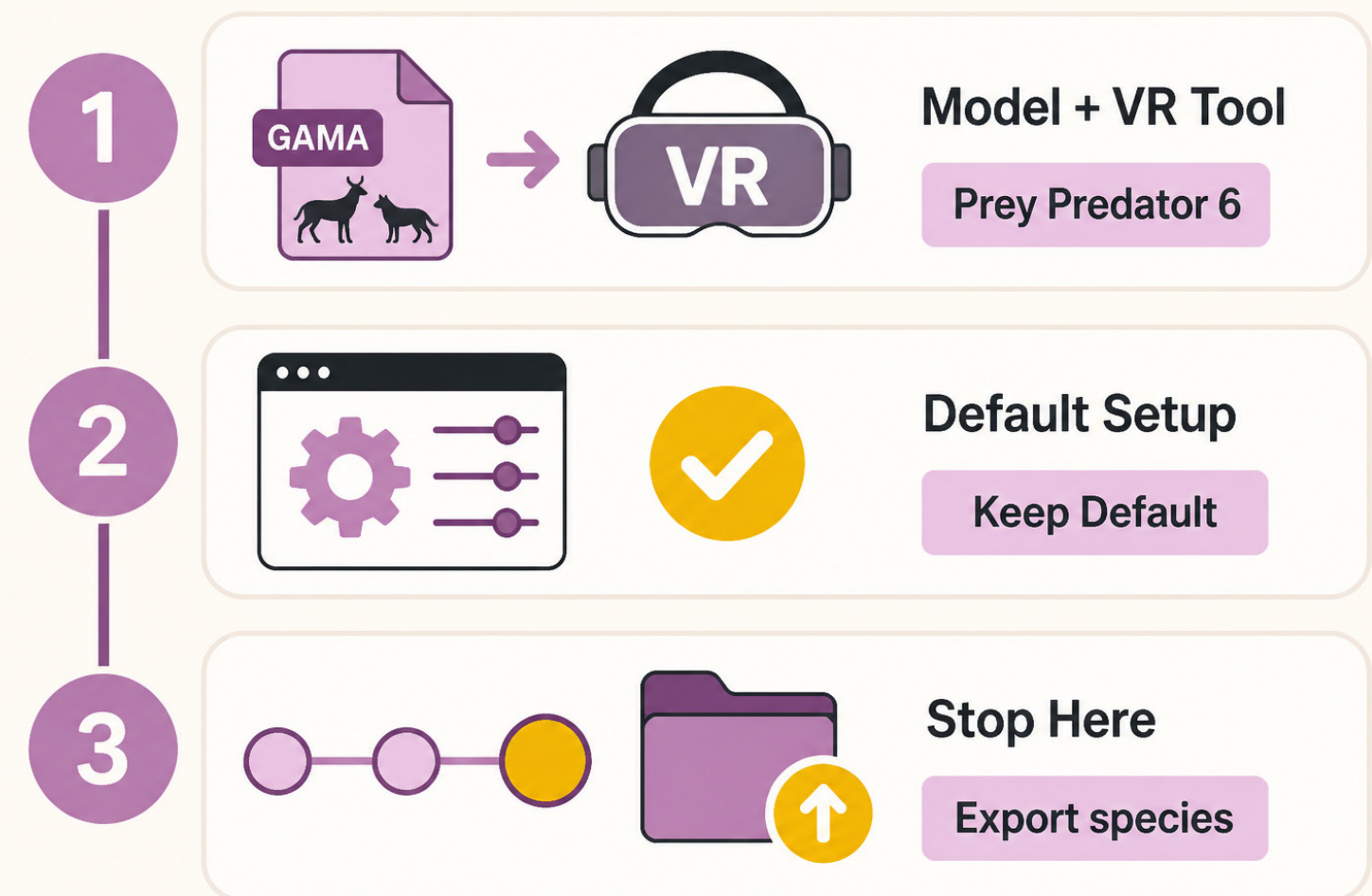
- Open the Prey Predator 6 model in GAMA.
- Start the SIMPLE VR experiment generator.

Keep the default setup

- Keep the default general values.
- Click Next to continue.

Stop at the right step

- Stop when you reach Export species.



REMINDER

GAMA setup

- Install the GAMA Plugin. 
- Open a model in the GAMA workspace and convert it into a Unity-compatible **vr_xp**. 

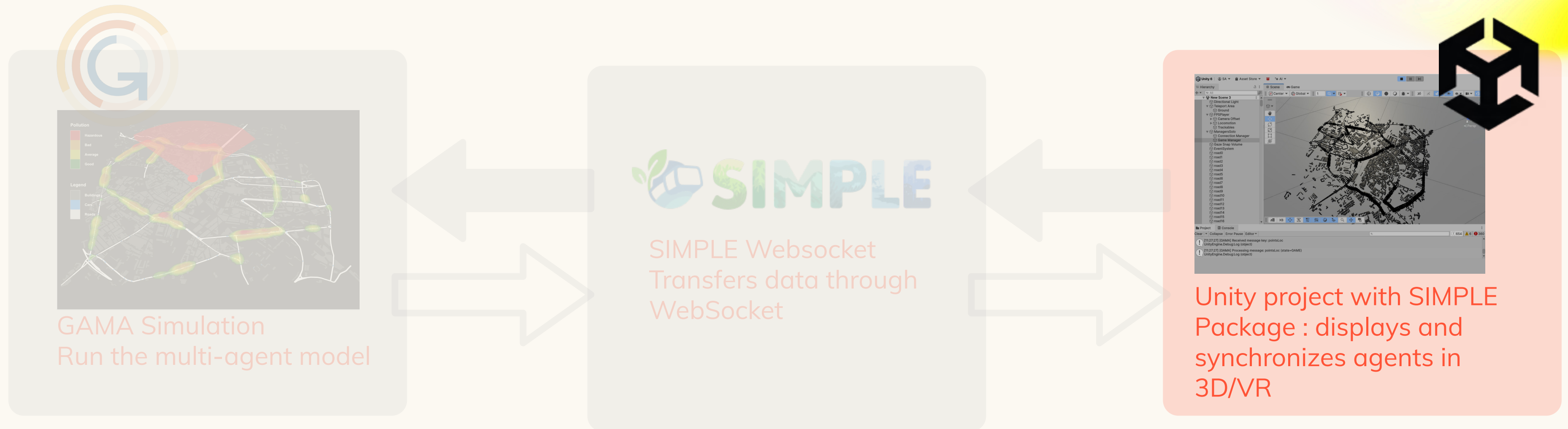
Unity setup

- Install and launch Unity 6000.0+
- import the package
- setup the scene

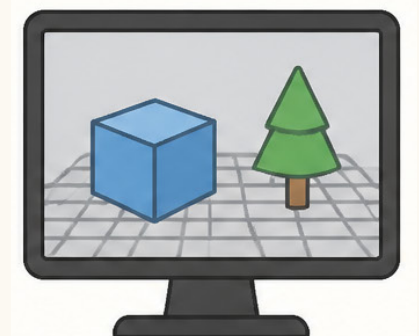
Middleware setup

- clone **simple.webplatform**
- Start the middleware with the **executable** in the folder.

Unity Package and Scene Setup



Live Build



Unity and Git

Required software

- **Unity 6** — version 6000.0 or later;
- recommended and tested version: Unity 6000.3.2f1;



Unity 6000.0 +



Git (Bash/Desktop)



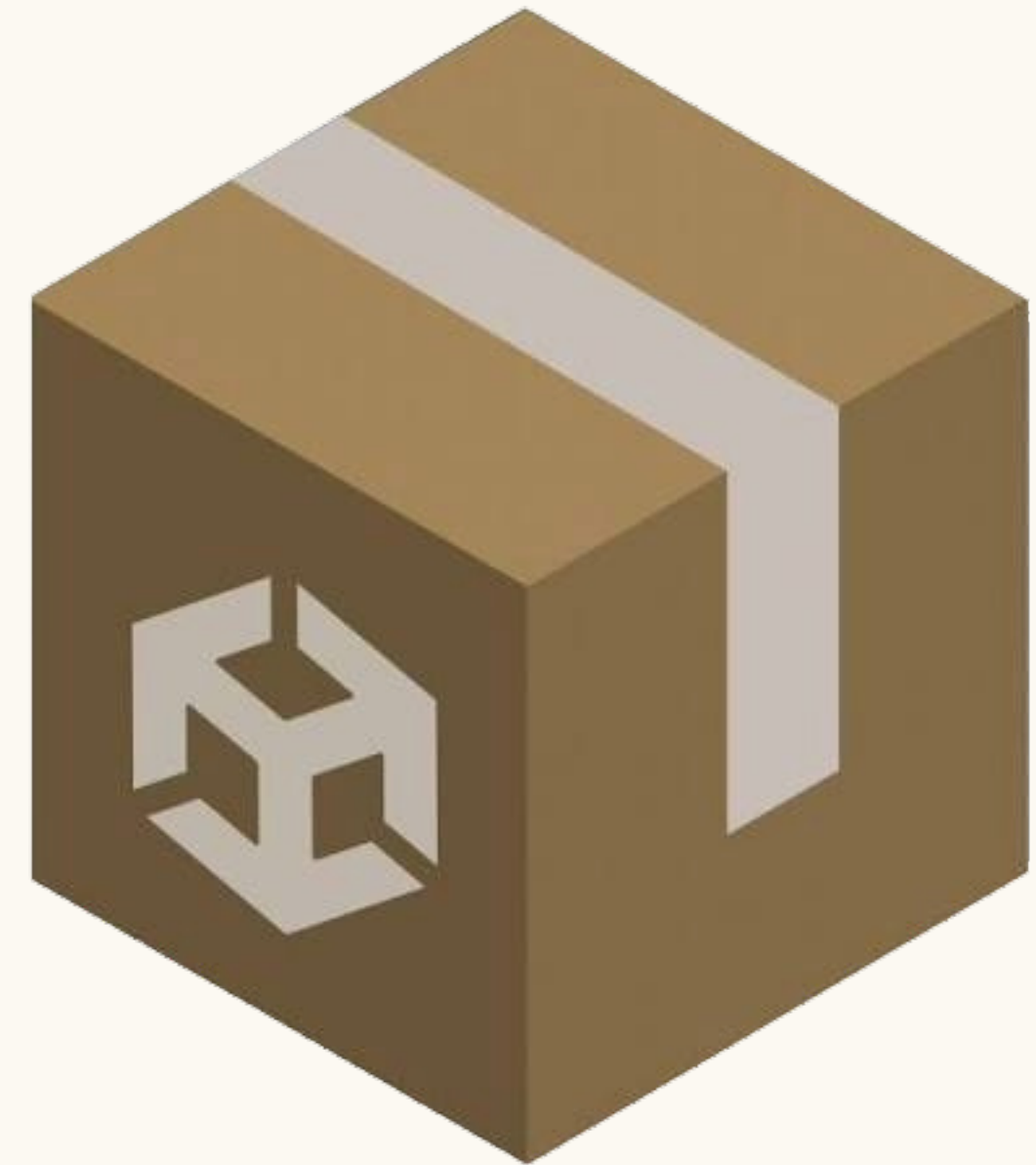
INSTALL THE UNITY PACKAGE

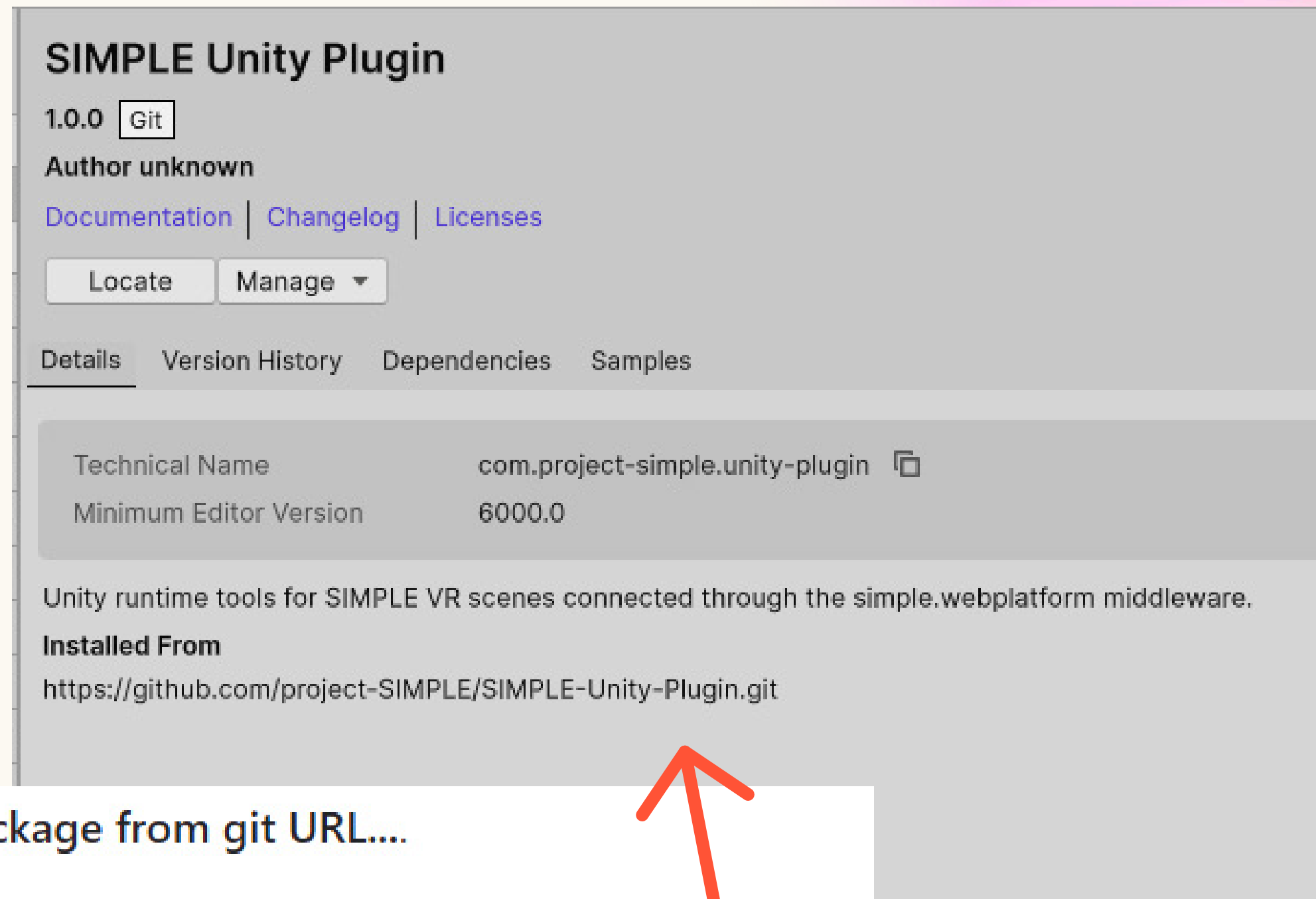
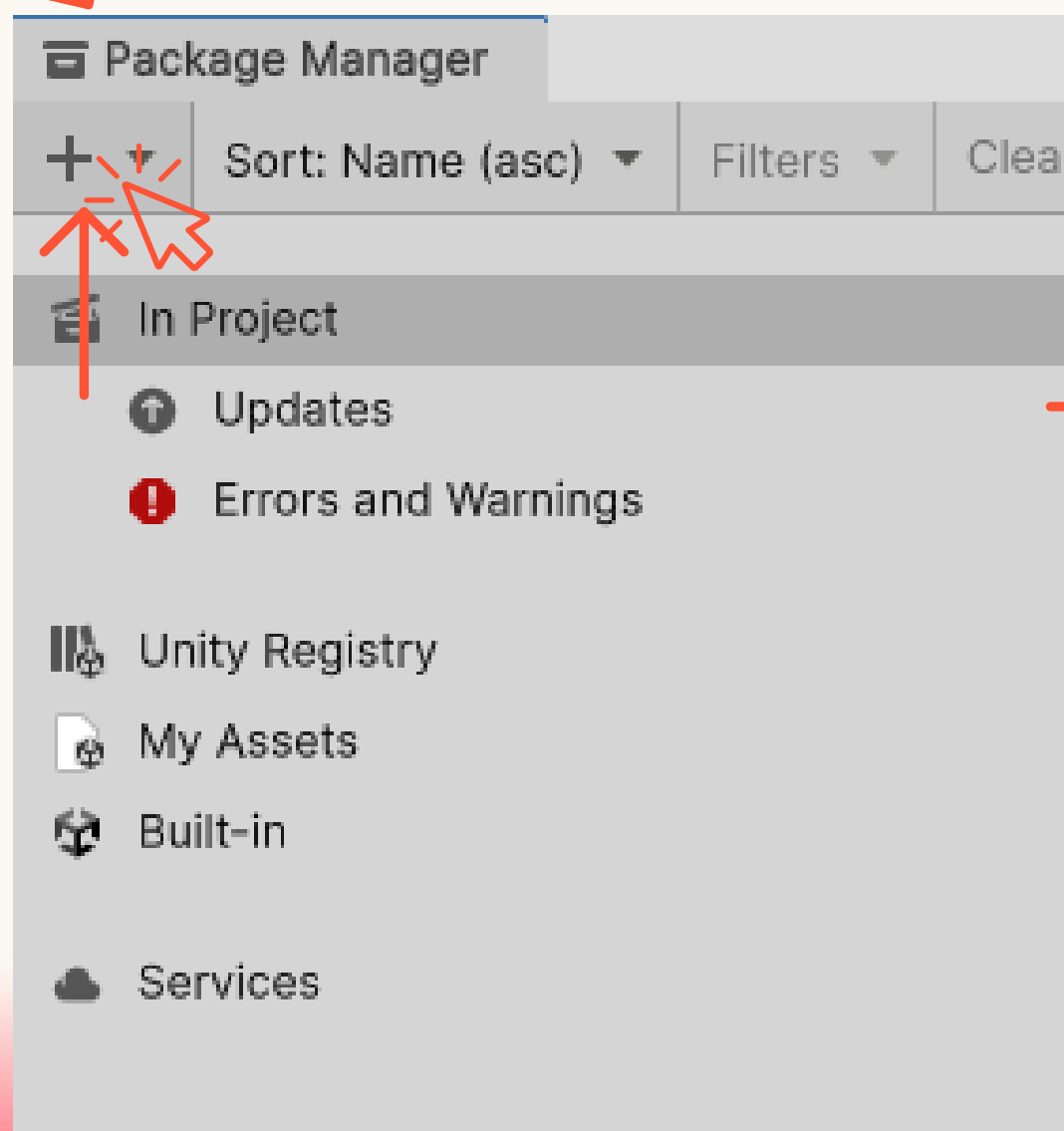
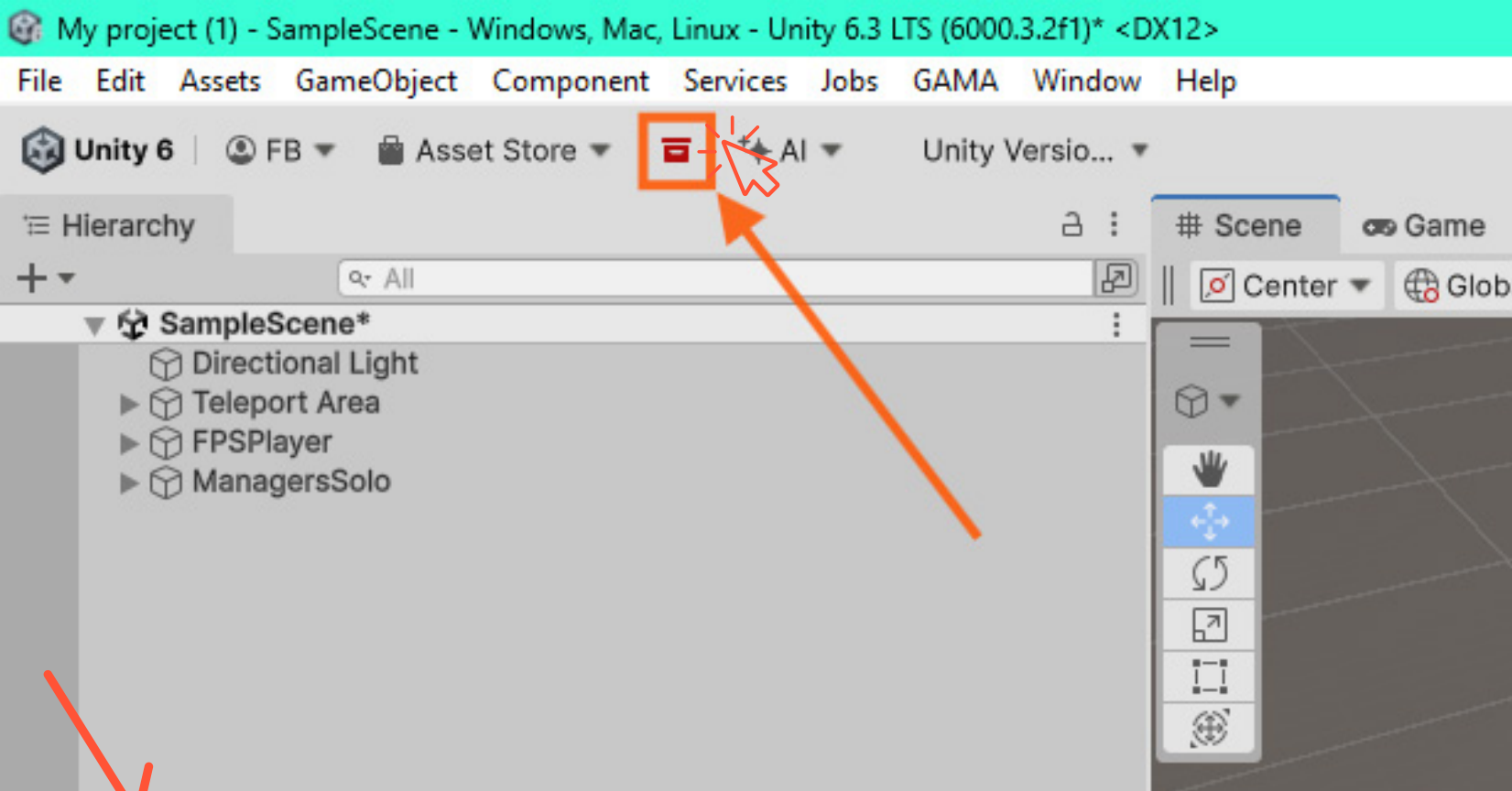
Install the Unity package

- Open Package Manager, then click +.
- Select Add package from git URL...
- Paste: <https://github.com/project-SIMPLE/SIMPLE-Unity-Plugin.git>

Local fallback (if needed)

- Use Add package from disk...
- Select package.json.





3. Select **Add package from git URL....**

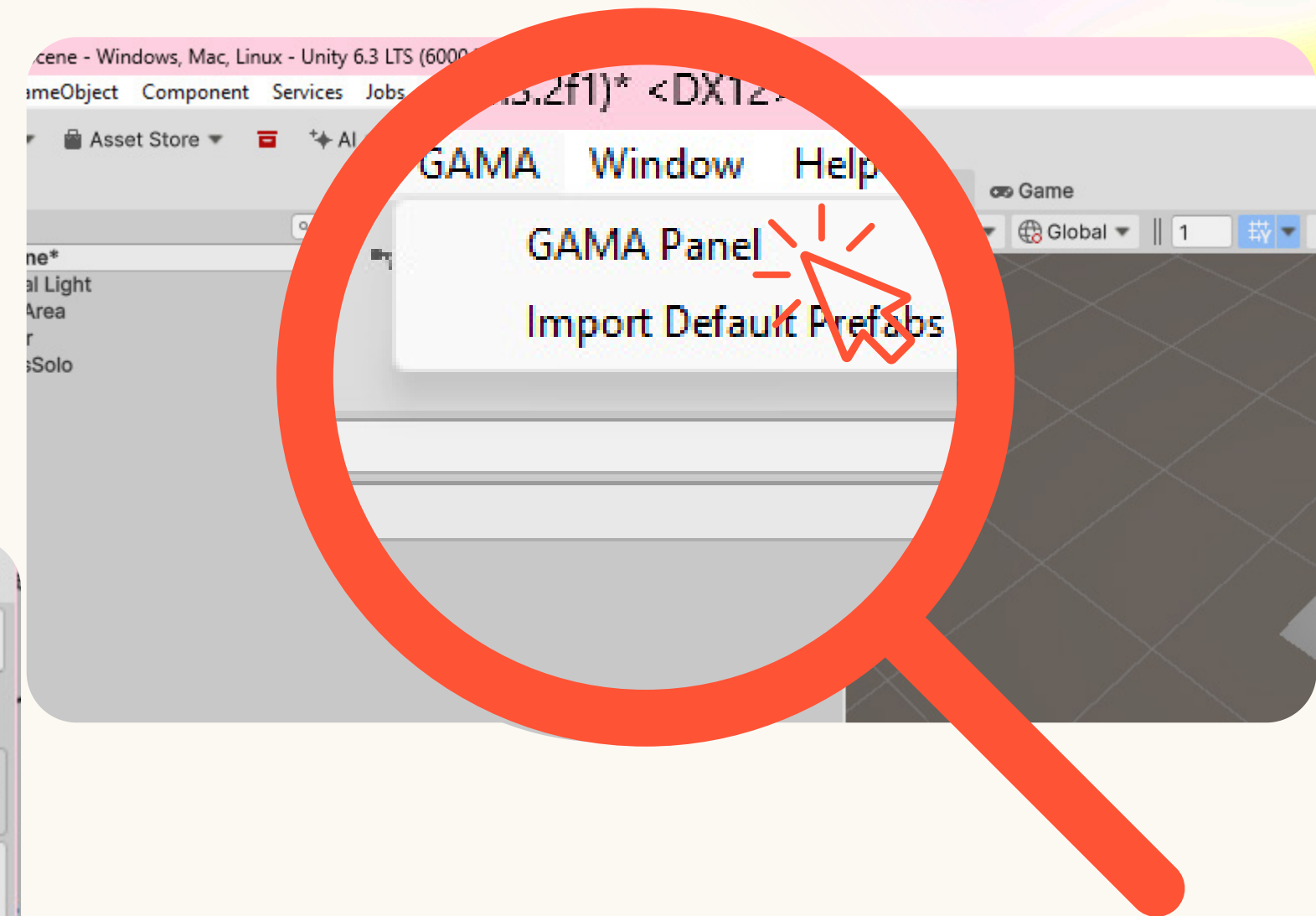
4. Enter:

```
https://github.com/project-SIMPLE/SIMPLE-Unity-Plugin.git
```

SET UP THE UNITY SCENE

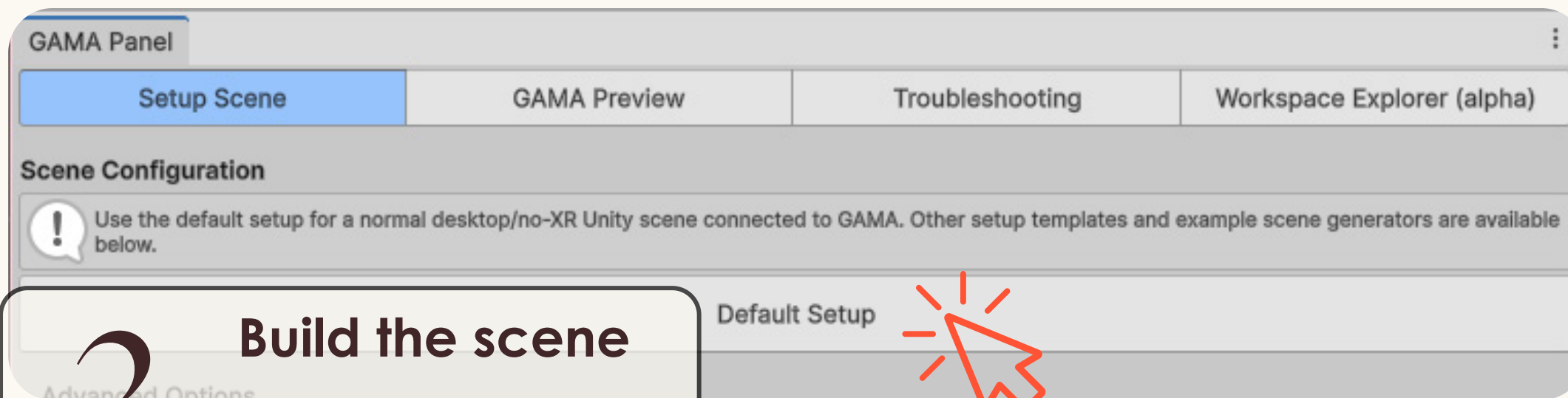
1. Open the setup tool

- In Unity, open: GAMA > GAMA Panel.



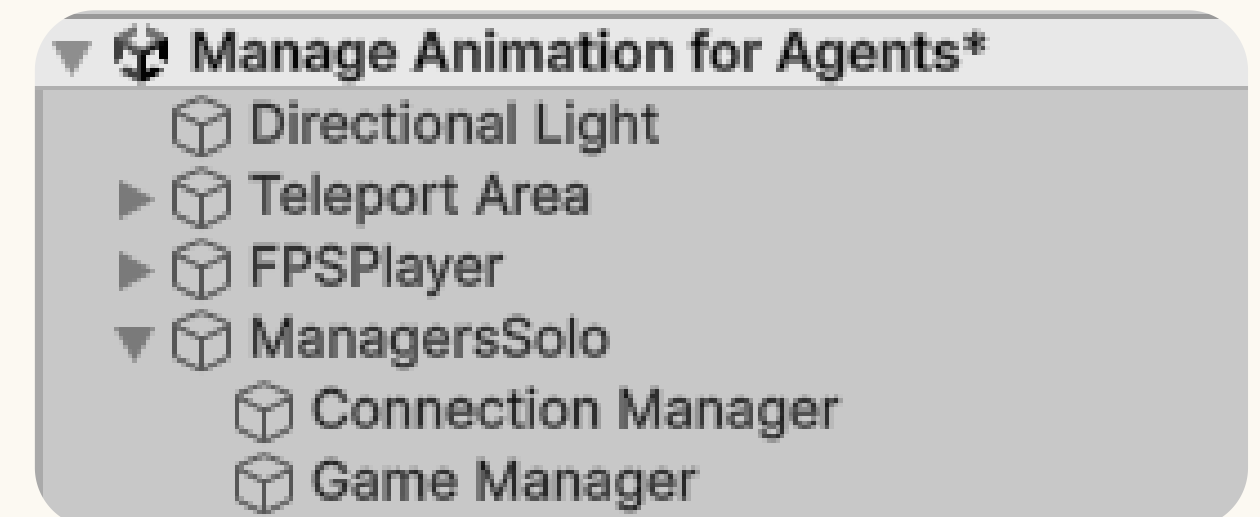
2. Build the scene

- Click **Setup Scene**



3. Verify the Hierarchy

- Confirm a player or camera rig exists in the scene.
- Confirm **Connection Manager** and **Game Manager** exist.
- Confirm preview and runtime roots are present.



REMINDER

GAMA setup

- Install the GAMA Plugin. ✓
- Open a model in the GAMA workspace and convert it into a Unity-compatible **vr_xp**. ✓

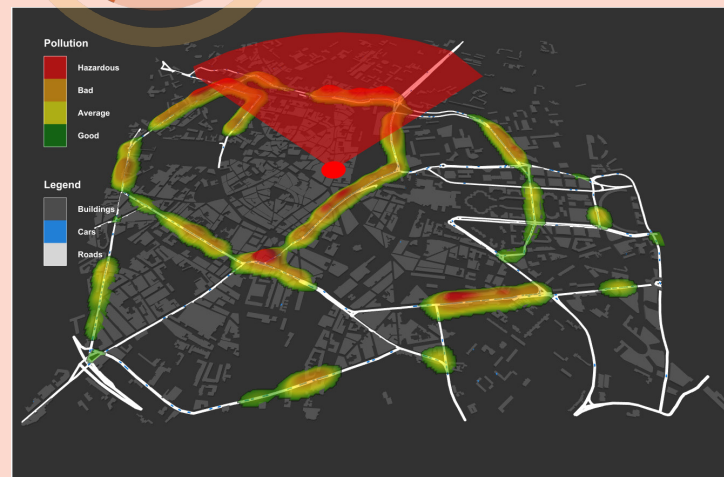
Unity setup

- Install and launch Unity 6000.0+ ✓
- import the package ✓
- setup the scene ✓

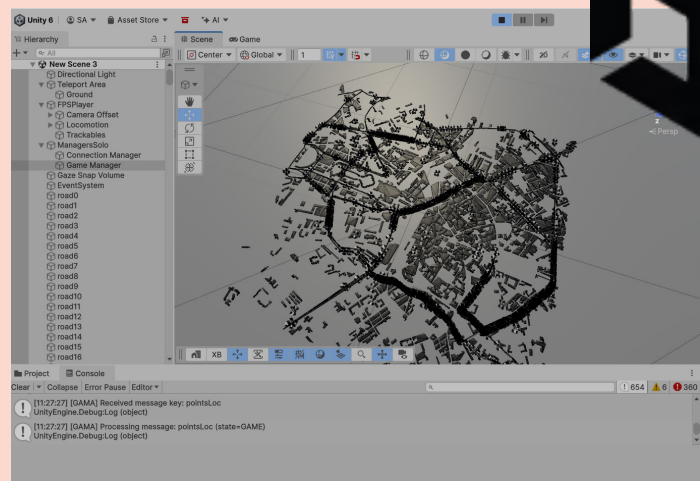
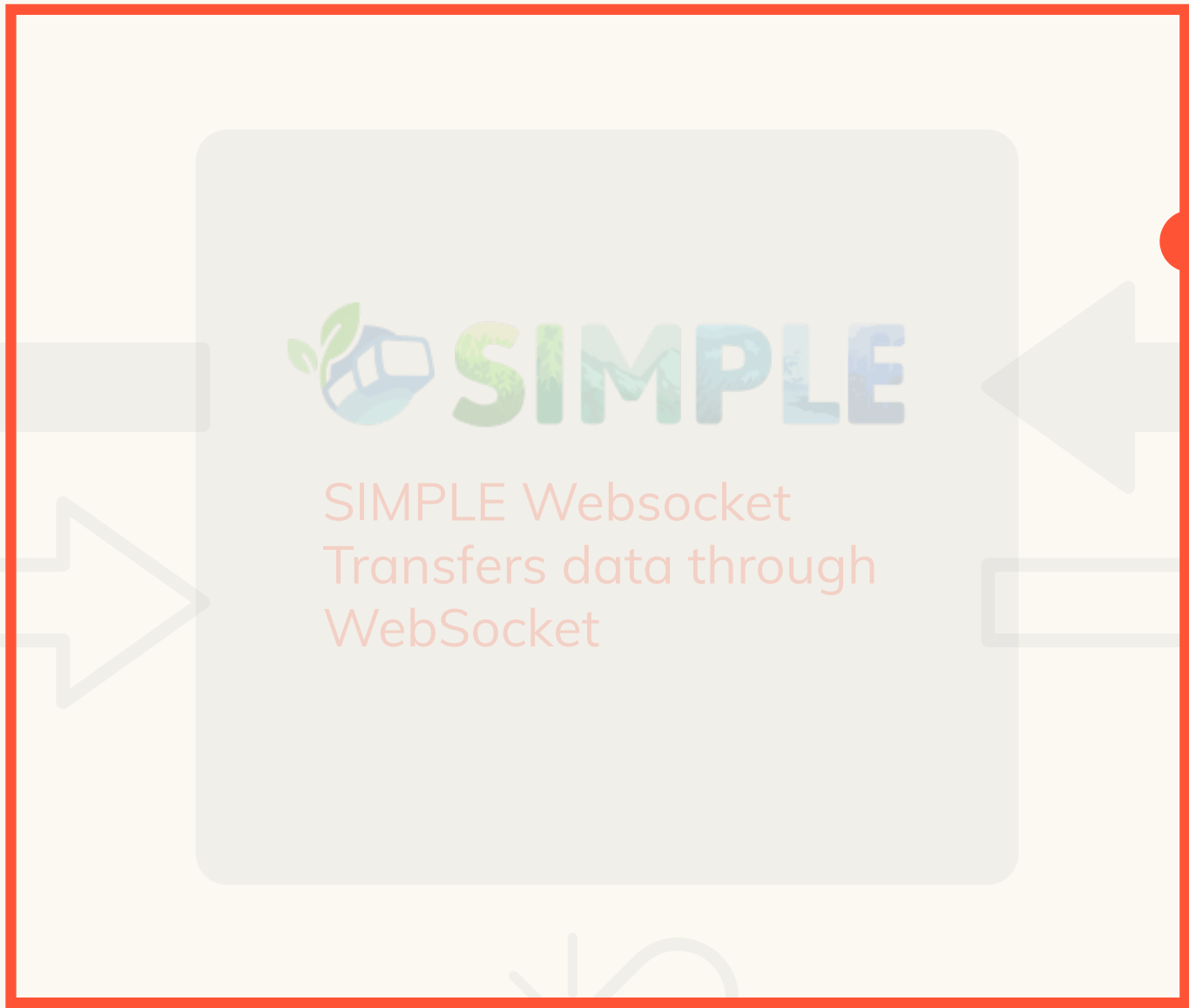
Middleware setup

- clone **simple.webplatform**
- Start the middleware with the **executable** in the folder.

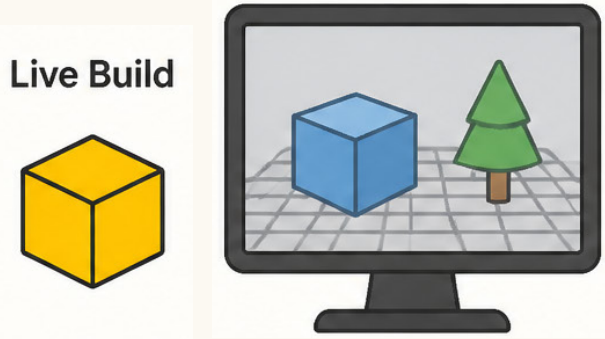
Middleware Setup



GAMA Simulation
Run the multi-agent model



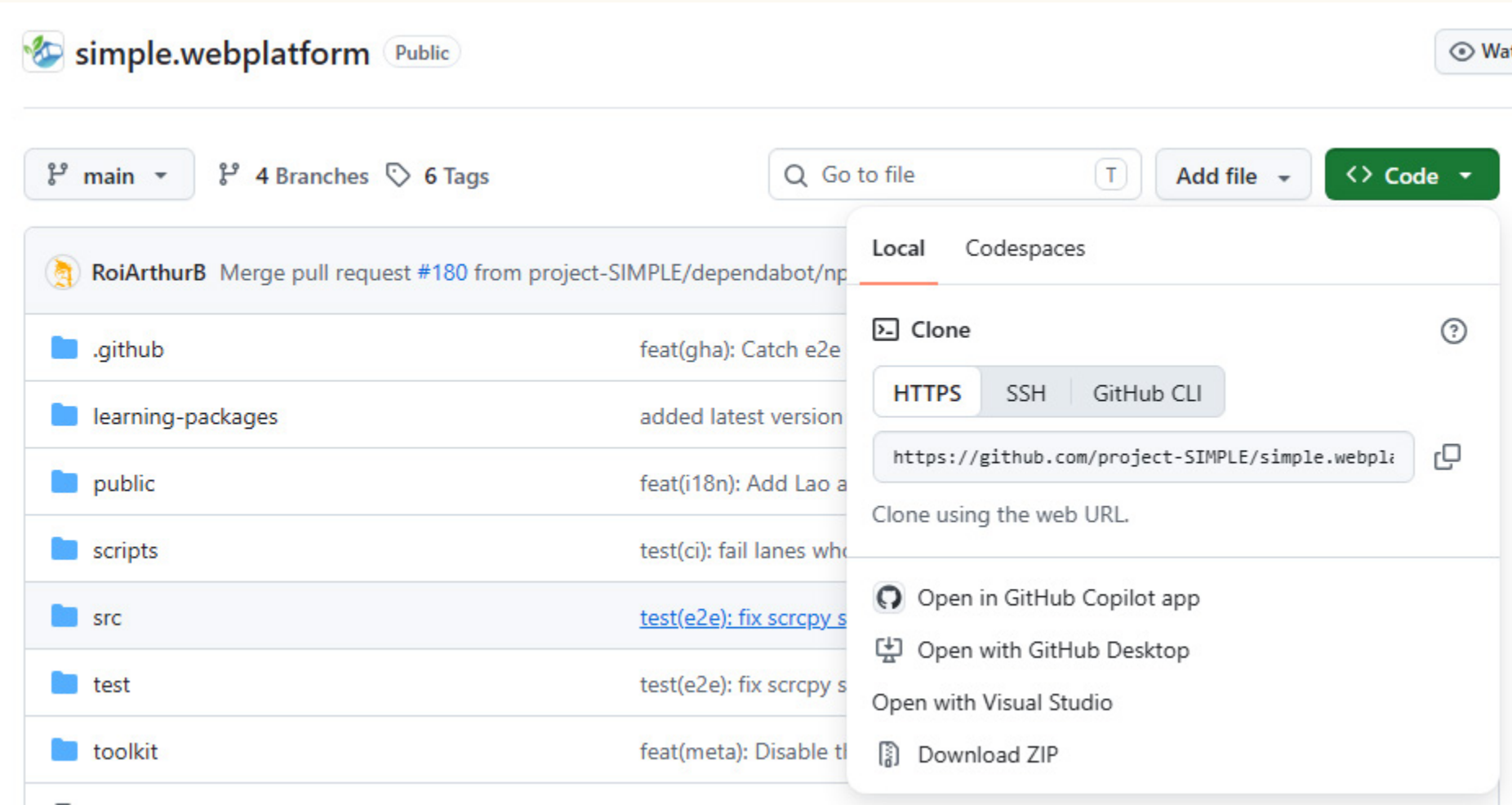
Unity project with SIMPLE
Package : displays and
synchronizes agents in
3D/VR



The Middleware

Required project files

- Clone **simple.webplatform** from the following repository : <https://github.com/project-SIMPLE/simple.webplatform.git>
- Download the **executable** at : <https://github.com/project-SIMPLE/simple.webplatform/releases/tag/v4.0>



The Middleware documentation : <https://github.com/project-SIMPLE/simple.webplatform>

The complete tutorial : <https://github.com/project-SIMPLE/SIMPLE-Unity-Plugin/tree/main/Documentation~/tutorial>

 main

 4 Branches

 6 Tags

 Add file

 Code

 **RoiArthurB** Merge pull request #180 from project-SIMPLE/dependabot/npm_and_... 1d116cf · 2 days ago 1,098 Commits

 .github	feat(gha): Catch e2e results for linux	3 days ago
 learning-packages	added latest version of quangbinhproject to git ignore	10 months ago
 public	feat(i18n): Add Lao and Khmer translation	3 weeks ago
 scripts	test(ci): fail lanes whose tests silently skipped, cover ModelM...	last week
 src	test(e2e): fix scrpy stream initialization flake in full-e2e by pr...	last week
 test	test(e2e): fix scrpy stream initialization flake in full-e2e by pr...	last week
 toolkit	feat(meta): Disable the Loft feature from the headsets	3 weeks ago
 .env.example	feat(env): Properly document and complete the example .env...	3 weeks ago
 .gitignore	test(ci): fail lanes whose tests silently skipped, cover ModelM...	last week
 .nvmrc	fix: Enforce binary compilation to use LTS 24	3 months ago
 BUILD_EXECUTABLE.md	feat: Bump Node to 24 + Ensure Vite is properly served from ...	6 months ago
 LICENSE	Create LICENSE	2 years ago

About

Server for monitoring Gama Server simulations and managing multiplayer connections

gama-platform

http

multiplayer-game

node-js

server

websocket

 Readme

 AGPL-3.0 license

 Activity

 Custom properties

 5 stars

 4 watching

 3 forks

Report repository

Releases 5

 Webplatform v4.0

2 days ago

Latest

+ 4 releases

<https://github.com/project-SIMPLE/simple.webplatform/releases/tag/v4.0>



▼ Assets 5

 simple-linux-v4	sha256:0c4a5303c4f03a0be8a5f8d9c6d86...		148 MB	2 days ago
 simple-macos-v4	sha256:71a471c87400c18c3f0df0cbe...	 2 	143 MB	2 days ago
 simple-win-v4.exe	sha256:6c7ba1f06c1e1fac74e3528905c8e...		120 MB	2 days ago
 Source code (zip)				2 days ago
 Source code (tar.gz)				2 days ago

folder	.git	06/08/2026 11:35	Dossier de fichiers	
folder	.github	13/05/2026 16:22	Dossier de fichiers	
folder	dist-api	17/06/2026 10:21	Dossier de fichiers	
folder	learning-packages	13/05/2026 16:22	Dossier de fichiers	
folder	node_modules	17/07/2026 15:32	Dossier de fichiers	
folder	public	13/05/2026 16:22	Dossier de fichiers	
folder	scripts	06/08/2026 11:35	Dossier de fichiers	
folder	src	06/08/2026 11:35	Dossier de fichiers	
folder	test	06/08/2026 11:35	Dossier de fichiers	
folder	toolkit	06/08/2026 11:35	Dossier de fichiers	
file	simple-win-v4.exe	06/08/2026 11:34	Application	123 180 Ko
file	errorLog.log	06/08/2026 11:37	Document texte	40 173 Ko
file	LICENSE	13/05/2026 16:22	Fichier	35 Ko
file	errorLog.log.1	22/07/2026 14:58	Fichier 1	102 393 Ko
file	.env	23/07/2026 11:13	Fichier ENV	1 Ko
file	.env.example	06/08/2026 11:35	Fichier EXAMPLE	4 Ko

Download the **executable** and drag it into your newly cloned simple.webplatform folder

First launch of the Middleware by double-clicking on the executable :

```
[simple] First launch: extracting runtime, please wait...
info    core·index      Starting the SIMPLE Webplatform !
info    infra·StaticServer =====

                                Webplatform started and is accessible
                                http://localhost:5173

                                =====
info    monitor·MonitorServ... Creating monitor server on: ws://0.0.0.0:8001
info    multi·PlayerManager  Creating monitor server on: ws://0.0.0.0:8080
warning simulation·ModelMan... Couldn't find settings file in folder C:\User
warning core·Controller      Couldn't find ADB working or started, cancell
info    core·Controller      Not running on M2L2 (Mac Mini) hardware – ski
info    core·Controller      Session timer started – shutdown sequence arm
```

if GAMA is closed

```
warning sim·GamaConnector    Reconnecting in 5s...
error   sim·GamaConnector    The platform can't connect to GAMA, please verify
000
warning sim·GamaConnector    Reconnecting in 5s...
info    sim·GamaConnector    Connected to Gama Server on ws://localhost:1000
```

if GAMA is open

REMINDER

GAMA setup

- Install the GAMA Plugin. ✓
- Open a model in the GAMA workspace and convert it into a Unity-compatible **vr_xp**. ✓

Unity setup

- Install and launch Unity 6000.0+ ✓
- import the package ✓
- setup the scene ✓

Middleware setup

- clone **simple.webplatform** ✓
- Start the middleware with the **executable** in the folder. ✓

Our connection is ready to be opened !



First Live Run and Verification

FIRST LIVE RUN

Launch order

- 1. Launch the Middleware
- 2. GAMA: open and run the vr_xp experiment.
- 3. Unity: press Play.
- 4. Wait for the live agents.

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Why the order matters

- Unity attaches to the experiment that is already active in GAMA.
- **Do not press Play in Unity until the WebSocket connection has been opened !!!**

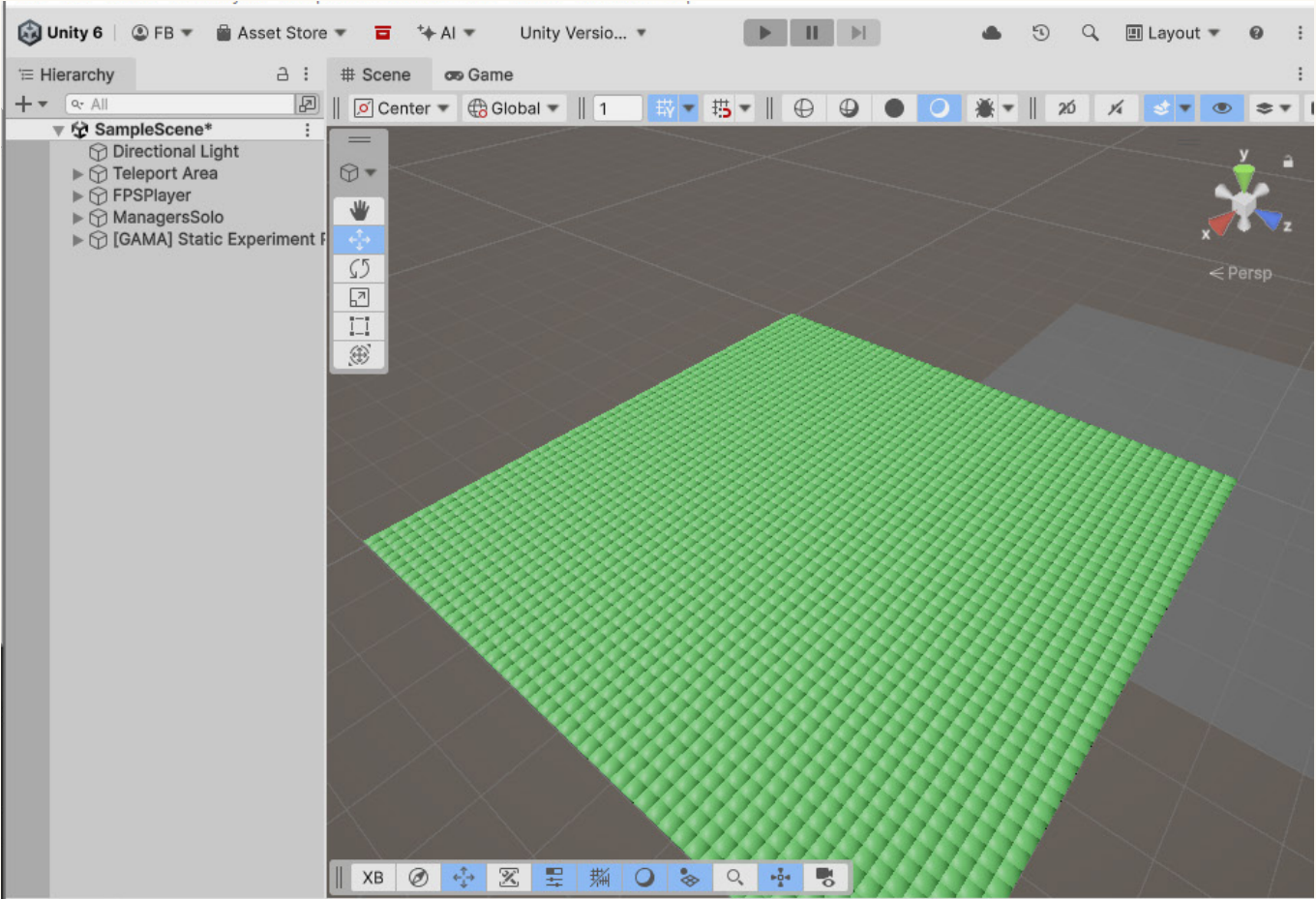
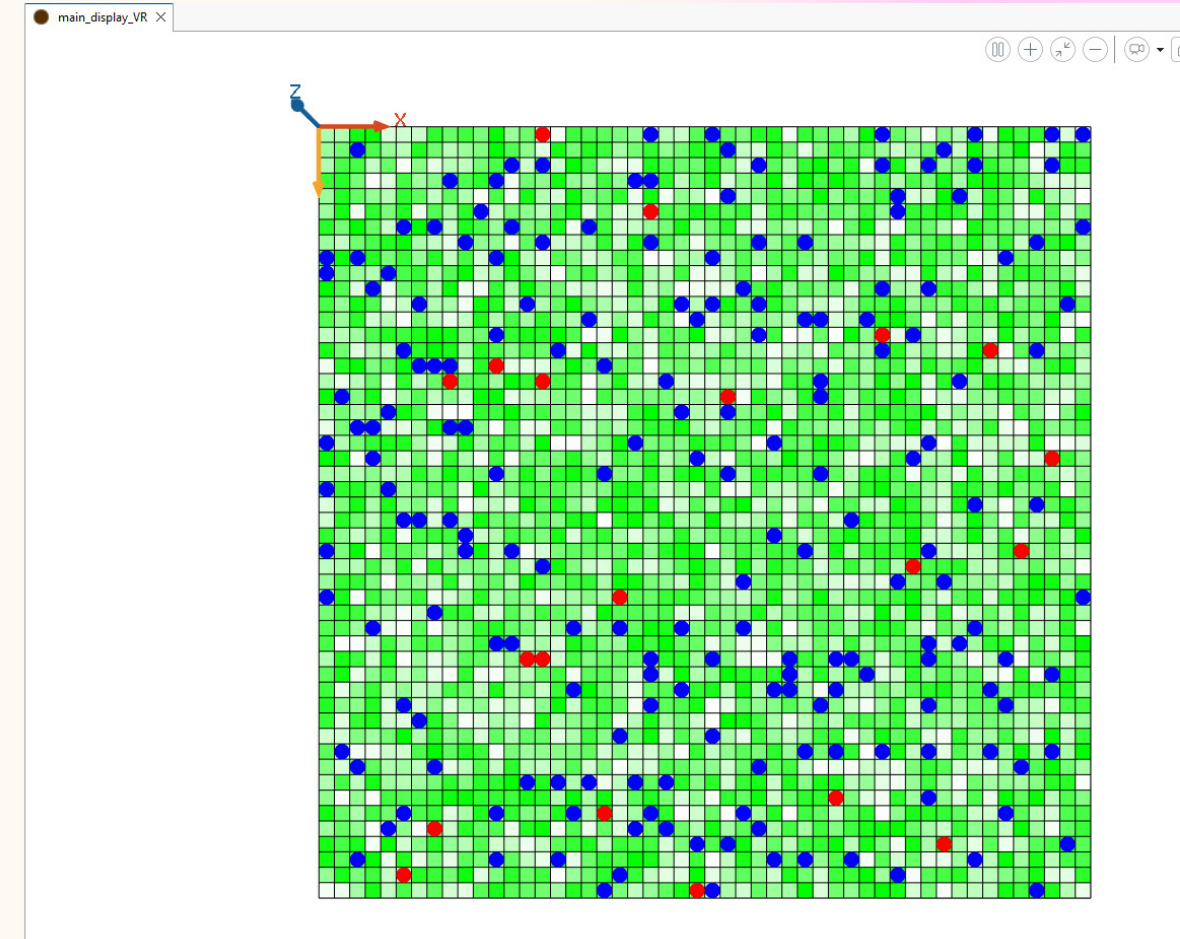
```
C:\Windows\system32\cmd.exe
Windows PowerShell
Copyright (C) Microsoft Corporation. Tous droits réservés.

Installez la dernière version de PowerShell pour de nouvelles fonctionnalités et améliorations ! https://aka.ms/PSWindows

PS C:\Users\flol\Desktop\simple.webplatform> npm start

> simple.webplatform@3.0.0 start
> concurrently "npx tsx src/api/index.ts" "vite"

(node:29888) [DEP0060] DeprecationWarning: The 'util._extend' API is deprecated. Please use Object.assign() instead.
(Use 'node --trace-deprecation ...' to show where the warning was created)
[1]
[1] VITE v7.3.3 ready in 518 ms
[1]
[1] → Local: http://localhost:8000/
[1] → Network: http://192.168.64.20:8000/
[0] info core:index Starting the SIMPLE Webplatform !
[0] warning simulation:ModelMan... Couldn't find settings file in folder C:\Users\flol\Desktop\simple.webplatform\learning
[0] warning core:Controller Couldn't find ADB working or started, cancelling ADB management
[0] warning infra:UpsManager UPS connection attempt 1/4 failed: 'No APC UPS found (vendorId=0x51d, productId=0x2). Is it
[0] info monitor:MonitorServ... Creating monitor server on: ws://0.0.0.0:8001
[0] info multi:PlayerManager Creating monitor server on: ws://0.0.0.0:8000
[0] info sim:GamaConnector Connected to Gama Server on ws://localhost:1000
[0] warning infra:UpsManager UPS connection attempt 2/4 failed: 'No APC UPS found (vendorId=0x51d, productId=0x2). Is it
[0] warning infra:UpsManager UPS connection attempt 3/4 failed: 'No APC UPS found (vendorId=0x51d, productId=0x2). Is it
[0] warning infra:UpsManager UPS connection attempt 4/4 failed: 'No APC UPS found (vendorId=0x51d, productId=0x2). Is it
[0] info core:Controller Session timer started - shutdown sequence armed for 3h if on battery
[0] warning infra:UpsManager Could not connect to UPS after 4 attempts - UPS monitoring disabled
```



```
C:\Users\flole\Desktop\simple × + ~
info core·index Starting the SIMPLE Webplatform !
info infra·StaticServer =====

Webplatform started and is accessible
http://localhost:5173

=====
info monitor·MonitorServ... Creating monitor server on: ws://0.0.0.0:8001
info multi·PlayerManager Creating monitor server on: ws://0.0.0.0:8080
warning simulation·ModelMan... Couldn't find settings file in folder C:\Users\flole\Desktop\simple.webplatform\learning-packages
warning core·Controller Couldn't find ADB working or started, cancelling ADB management
info core·Controller Not running on M2L2 (Mac Mini) hardware – skipping UPS connection
info core·Controller Session timer started – shutdown sequence armed for 3h if on battery
info sim·GamaConnector Connected to Gama Server on ws://localhost:1000
warning multi·PlayerManager Can't find player with ID unity_play_20260720043442_423015ff
error multi·PlayerManager Cannot find player with ID unity_play_20260720043442_423015ff}
error multi·PlayerManager Something strange happened while try to change in_game status for unity_play_20260720043442_423015ff false
error multi·PlayerManager Cannot find player with ID unity_play_20260720043442_423015ff}
warning multi·PlayerManager Can't remove un-existing player undefined
error monitor·MonitorServ... [ERR 1006] Message too big!
error monitor·MonitorServ... Message : ArrayBuffer {
  (detached),
  [byteLength]: 0
}
error monitor·MonitorServ... Message size: 45 bytes
error monitor·MonitorServ... [ERR 1006] Message too big!
error monitor·MonitorServ... Message : ArrayBuffer {
  (detached),
  [byteLength]: 0
}
error monitor·MonitorServ... Message size: 45 bytes
error monitor·MonitorServ... [ERR 1006] Message too big!
error monitor·MonitorServ... Message : ArrayBuffer {
  (detached),
  [byteLength]: 0
}
error monitor·MonitorServ... Message size: 45 bytes
info multi·PlayerManager New connection of the player of id unity_play_20260720043442_423015ff
warning multi·PlayerManager The last message received from unity_play_20260720043442_423015ff had an unknown type
{
  type: 'unity_diagnostics',
}
```

Launching the GAMA experiment :

Experiment Agents Views Support

Simulation 0: 0 cycle elapsed [00:00:00]

▶ ⏸

↺

↻

✖

predator_VR / Experiment vr_xp

main_display_VR

Idle

⏸ + ↺ −

🗨 🔒 📷 🔍

▶ prey_predator ▶ vr_xp

1 model prey_predator_model_VR

2

3 import "Model 06.gaml"

4

5 species unity_linker parent: abstract_unity_linker {

Interactive console

+

−

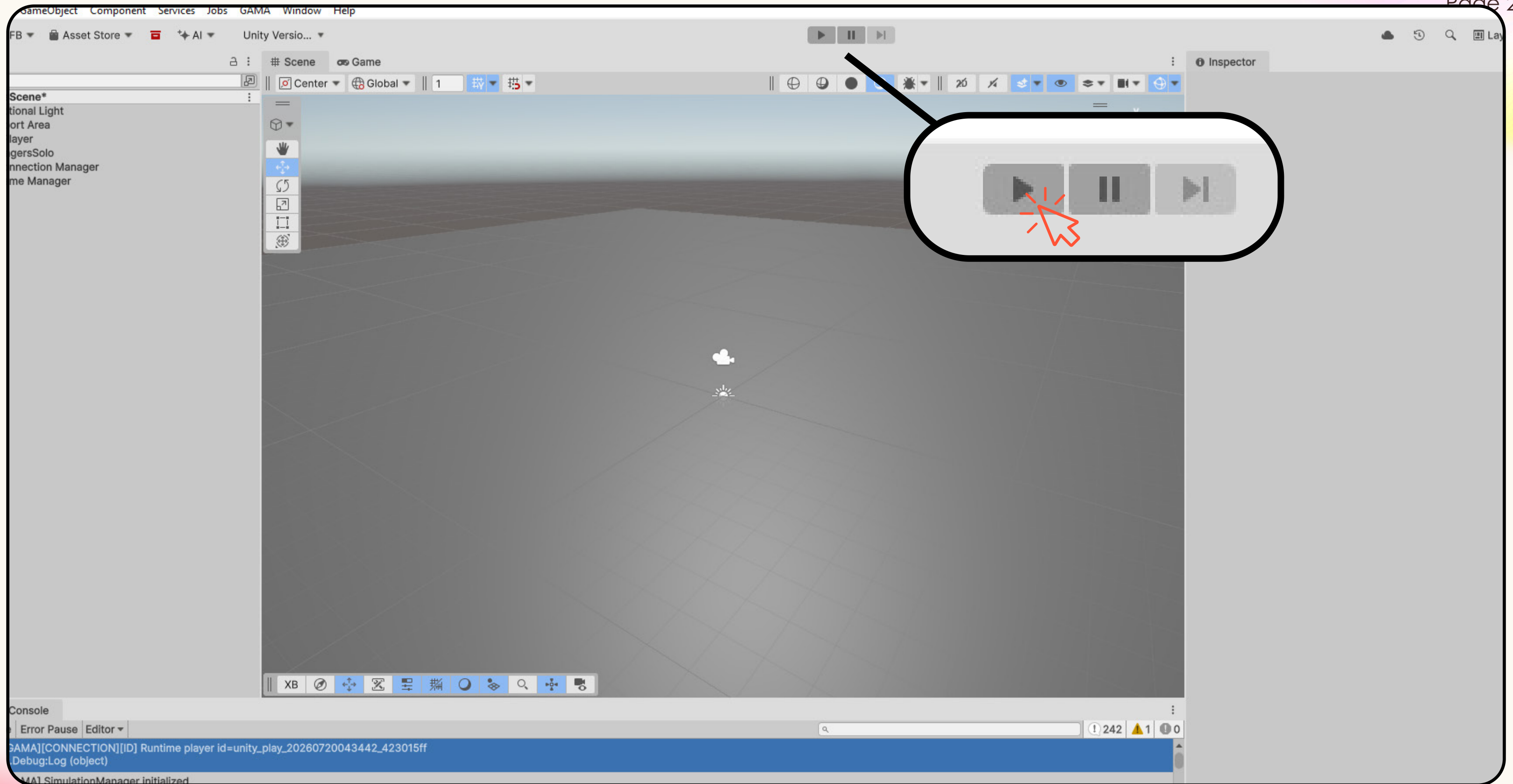
⏸

z

x

Launching the connection from Unity:

Page 24

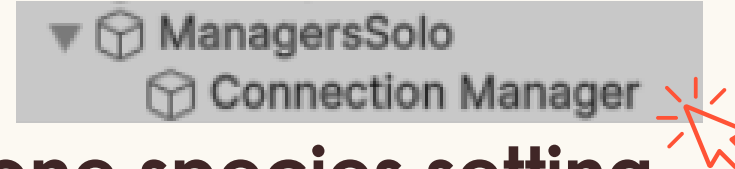


Visual Personalization

PERSONALIZE DURING PLAY MODE

Edit while Play Mode runs

- Keep Unity in Play Mode so the agents keep updating.
- Select **Game Manager** in the Hierarchy to access species controls.

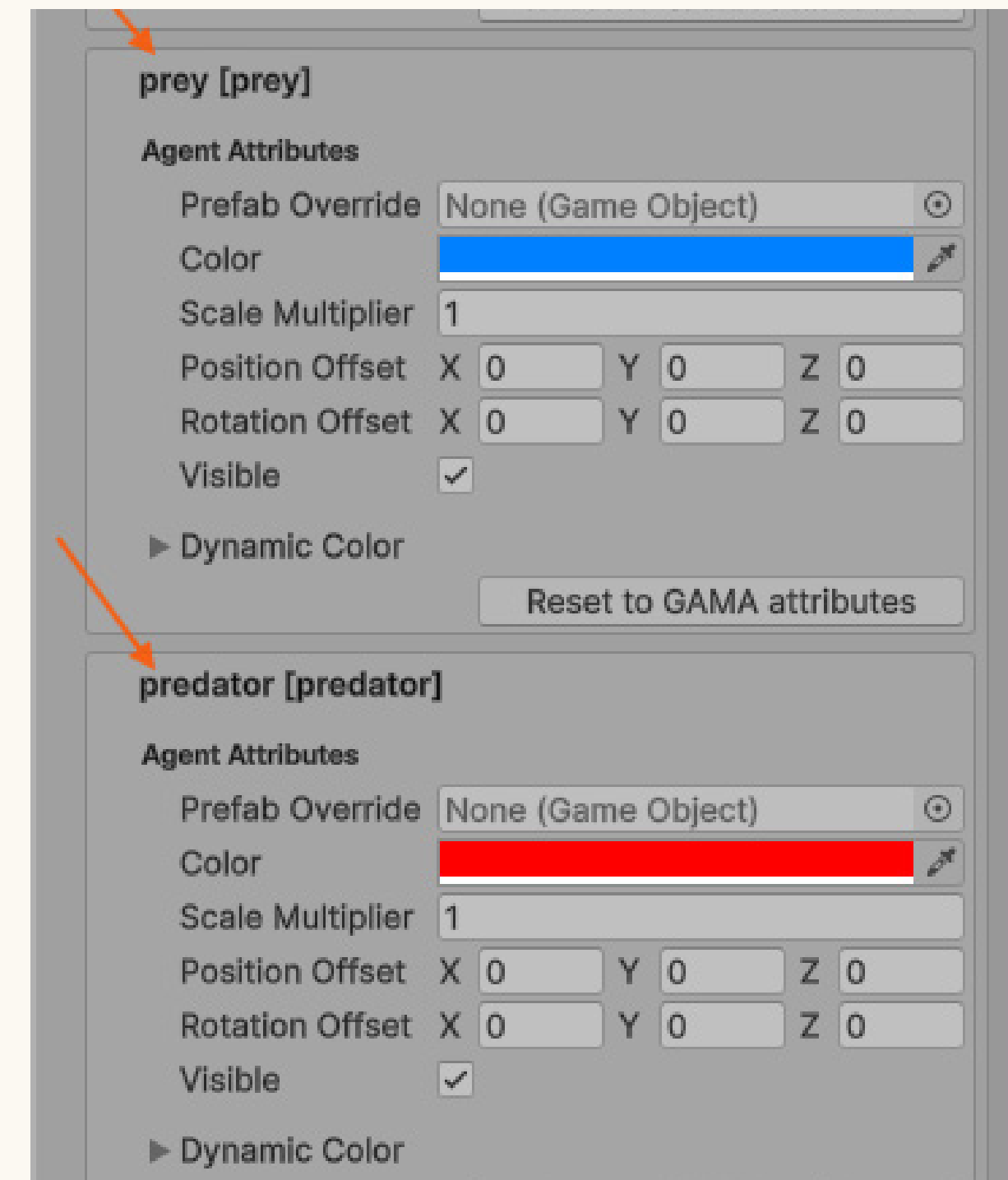


Change one species setting

- Open the species settings and pick one species to edit.
- Change one visual setting, then watch the scene update.

Various aspects

- Prefab
- Color
- Position & Rotation
- Visibility



YOUR TURN - MAKE THE LIVE SCENE READABLE

Color the species

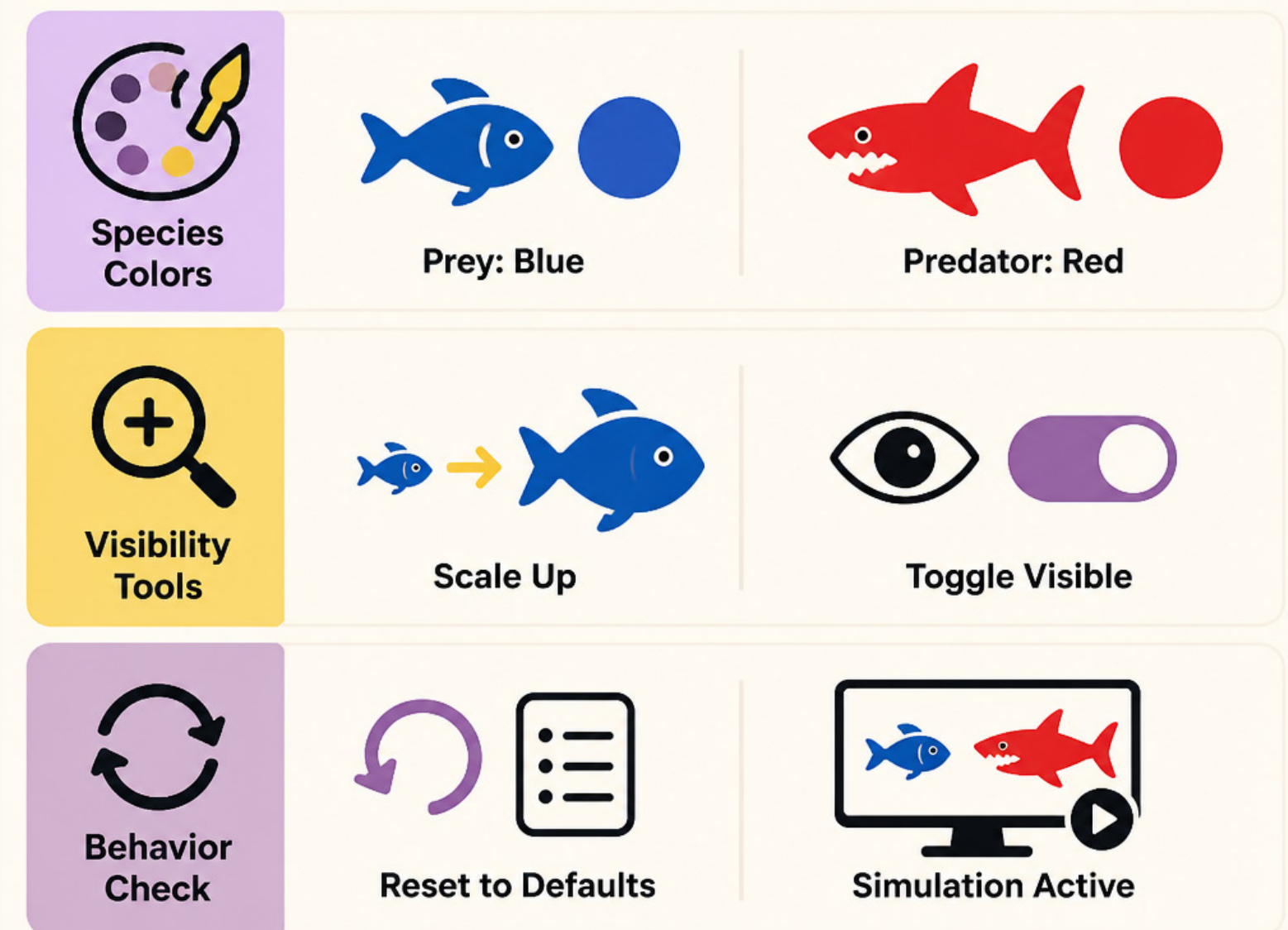
- Set prey to blue so it is easy to follow.
- Set predator to red so it stands out clearly.

Improve visibility

- Increase the scale for both species to see them better.
- Toggle Visible on one species to test hiding and showing.

Confirm live behavior

- Reset one local change to return to GAMA attributes.
- Check that the simulation keeps running while visuals update.



Static Preview Usage

GENERATE A PREVIEW

Open the preview tool

- Go to **GAMA > GAMA Panel > GAMA Preview** to access the preview controls.

Prepare GAMA and the Middleware

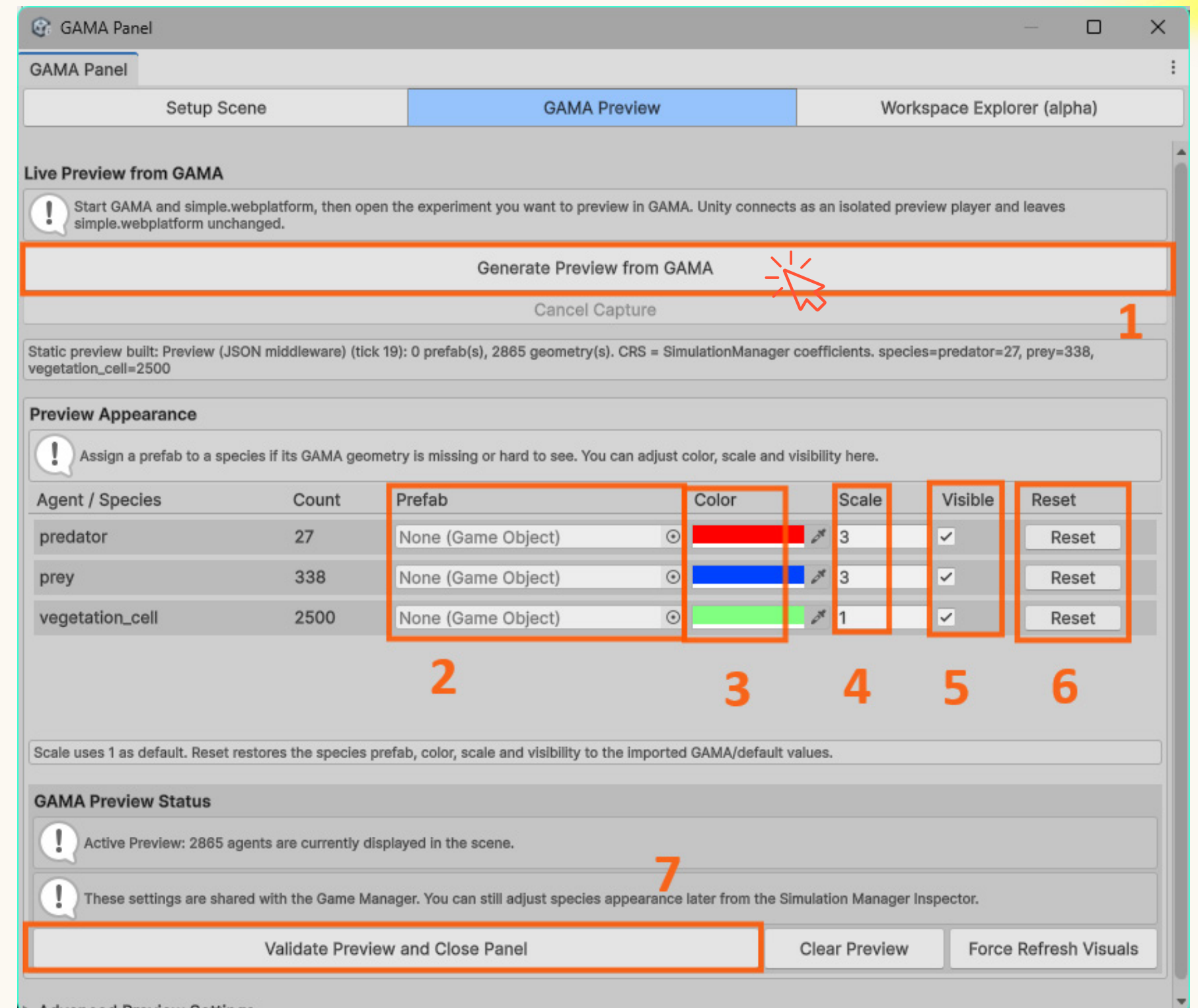
- Make sure the **Middleware** is launched
- Then, run the **GAMA experiment**

Generate the snapshot

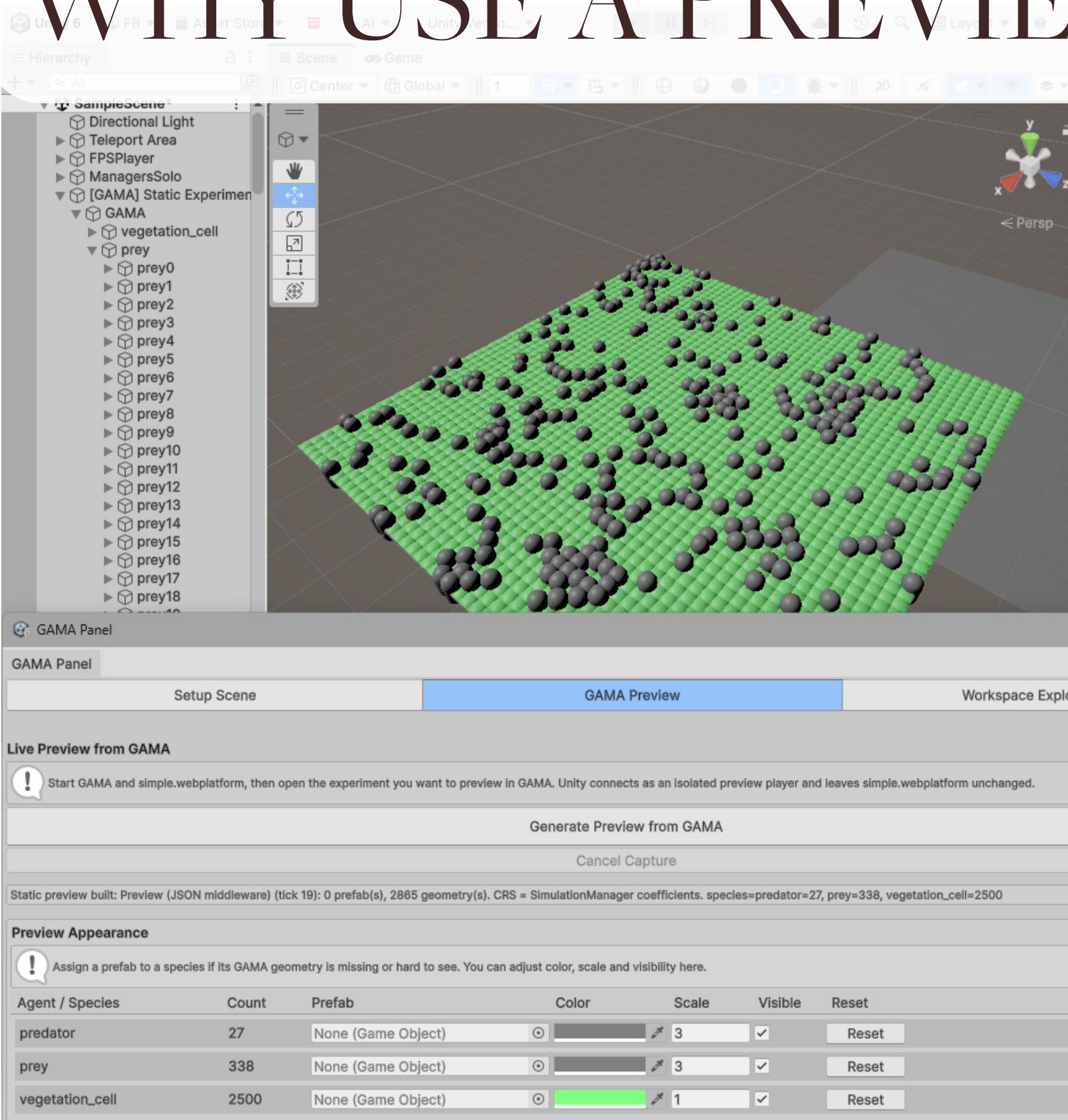
- Click **Generate Preview from GAMA** and wait...
- ... until the capture is complete before doing anything else.

Expected result

- The map and agents appear in Edit Mode as a static scene.



WHY USE A PREVIEW?



Why a preview helps

- Play Mode is live, but adjusting visuals there can feel slow.

Work faster in Edit Mode

- Use a static preview to inspect the scene without running time.
- Try prefabs, colors, scale, and visibility with quick feedback.

Prepare once, reuse later

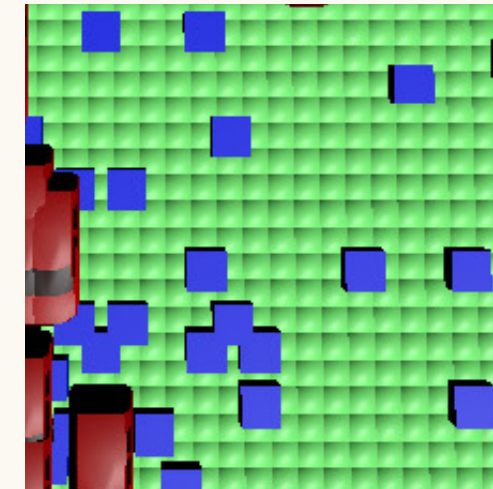
- Set up the scene before Play Mode, then reuse the same settings.
- Remember: preview is a **snapshot**, not a running simulation.

Dynamic Color Integration

STATIC COLOR OR DYNAMIC COLOR?

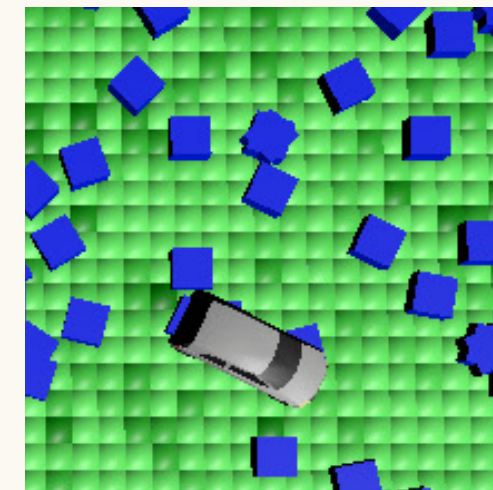
Static color

- All agents in one species share the same single color.
- Example: every predator stays red at all times.



Dynamic color

- Each agent's color depends on a value sent by GAMA.
- Example: vegetation color changes with the food attribute.



update color of each agent



GAMA

update "food" attribute



SIMPLE

Middleware

"food" attribute



Unity



We want **Unity** to catch **attributes** sent by **GAMA** to color the corresponding agents.

SEND AN ATTRIBUTE FROM GAMA

Add the food attribute to vegetation_cell

- Keep the existing species calls for prey and predator.
- Create a list<float> from vegetation_cell collect each.food.
- Build grass_atts with the key **"food"** for Unity.
- Send vegetation_cell with up_vegetation_cell and grass_atts.

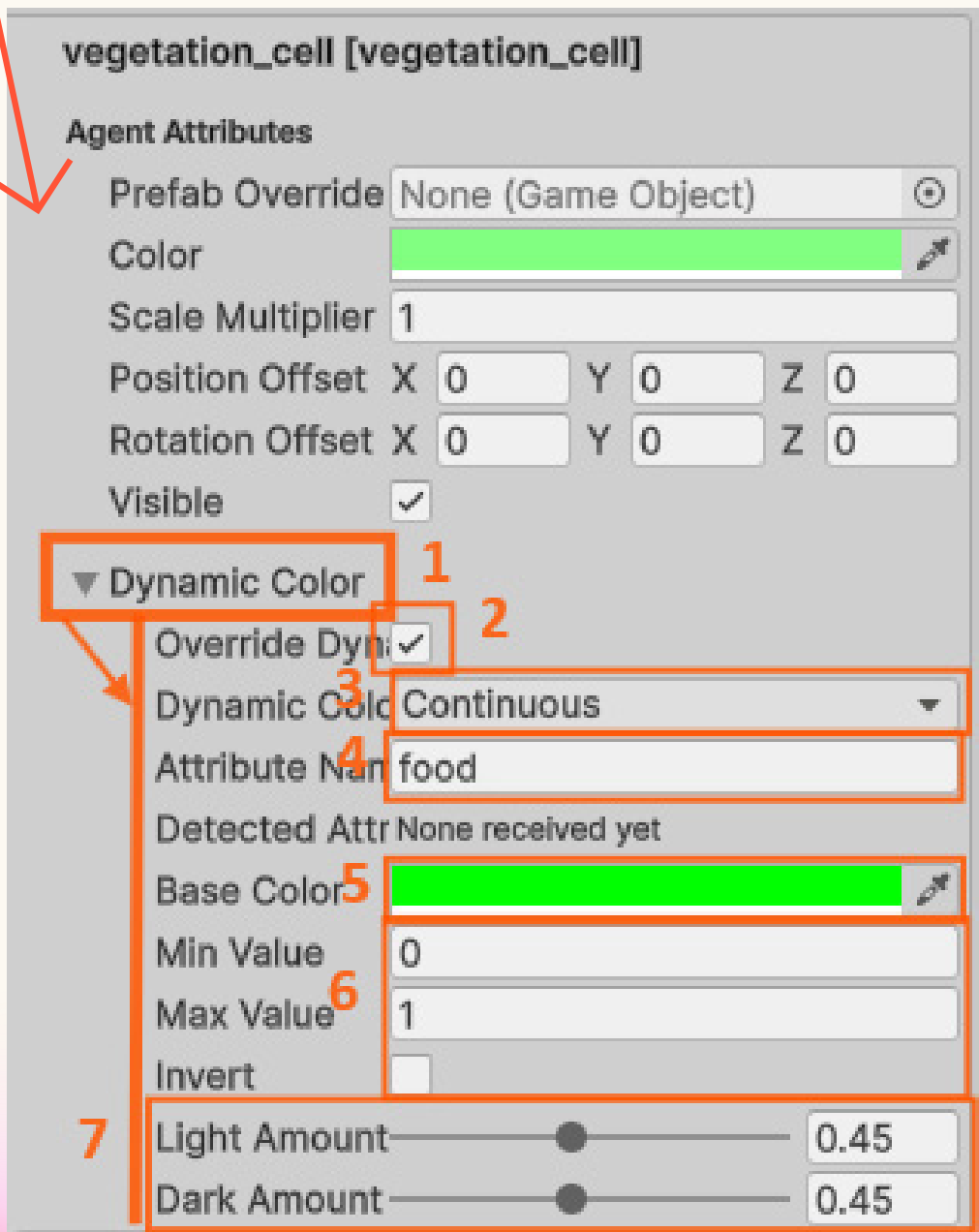
Code to paste (send_geometries)

```
reflex send_geometries { list<float> grass_food <- vegetation_cell collect each.food;  
map<string, list<float>> grass_atts <- ["food":: grass_food];  
do add_geometries_to_send(pre, up_pre);  
do add_geometries_to_send(predator, up_predator);  
do add_geometries_to_send(  
  vegetation_cell,  
  up_vegetation_cell,  
  grass_atts  
);  
do add_geometries_to_send(  
  generic_species,  
  up_generic_species  
);  
}
```



CATCH THE ATTRIBUTE IN UNITY

In Unity, select **Game Manager**, then **vegetation_cell**, then **Dynamic Color**



Configure Dynamic Color

- Enable Override Dynamic Color, then choose Continuous mode.
- Set Attribute Name to food and set Base Color to green.

Set the value range

- Set Min Value to 0 and set Max Value to 1.
- Set Light and Dark Amount to 0.5, then try Invert.



Afternoon Mission and Guided Lab